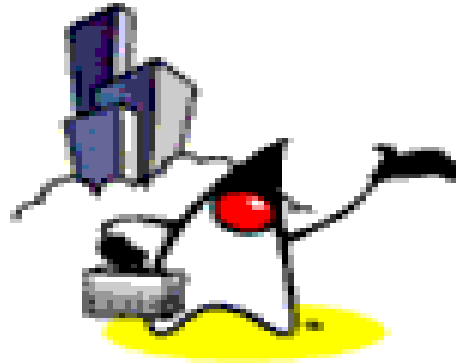


Java Array



Agenda

- What is an array?
- Declaration of an array type variable
- Instantiation of an array object
- Accessing array element within an array object
- Array length
- Multi-dimensional array



What is an Array?

Why you want to use an Array?

- Suppose we have three variables of **same type *int***

```
int number1;  
int number2;  
int number3;
```

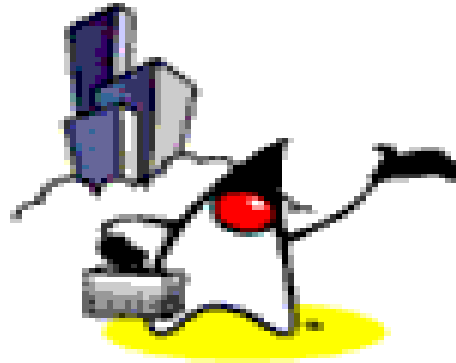
```
number1 = 1;  
number2 = 2;  
number3 = 3;
```

As you can see, initializing and using them individually is a tedious task. And it will get worse as the number of variables is getting larger.

Introduction to Arrays

- In Java (and other programming languages), there is a feature wherein we use one variable to store **multiple items of same type** and manipulate them. This type of variable is called an array.
- An array stores multiple data items of the same type, in a contiguous block of memory, divided into a number of slots.

	0	1	2
number:	1	2	3



Declaration of an Array

Declaring an Array Type variable

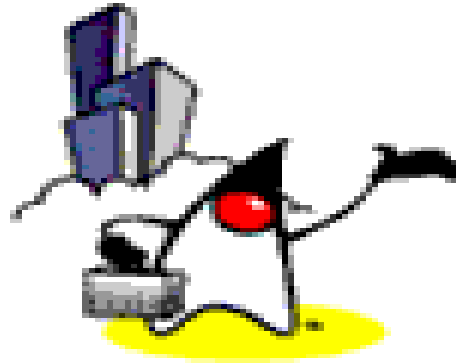
- To declare an array, write the data type, followed by square brackets [], followed by the variable name.

- For example,

```
int[] ages; // "ages" is int array type
```

or

```
int ages[]; // This is allowed as well
```



Instantiation of an Array Object

Array Object Instantiation

- After declaring array type variable, we can create an array object instance and specify its length with a constructor statement.
- What does “Object Instantiation” mean?
 - > In Java, this means creation of an object
- What is a “Constructor”?
 - > In order to instantiate an object, Java calls a constructor method (We will cover more about instantiating objects and constructors in detail later.)

Array Object Instantiation

- To instantiate (or create) an array object, use the `new` keyword, followed by the square brackets containing the number of items you want the array to have
 - > Array cannot be resized

- For example,

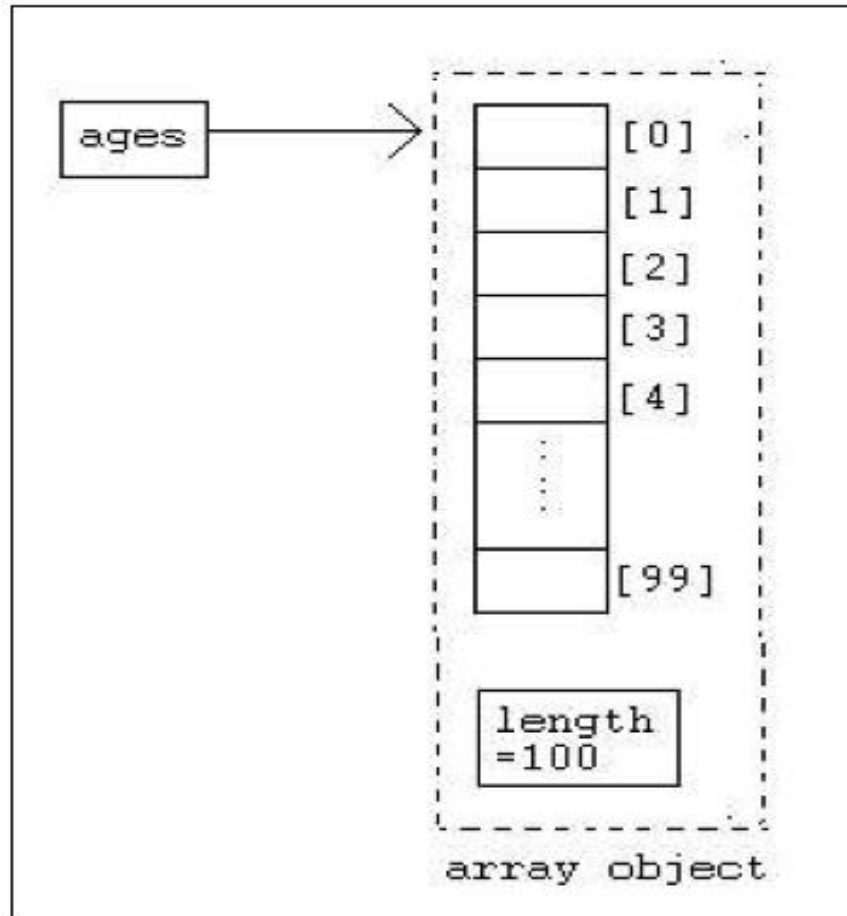
```
//declaration  
int[] ages;
```

```
//instantiate int array object with length of 100  
ages = new int[100];
```

or, can also be written as,

```
//declare and instantiate object  
int[] ages = new int[100];
```

How does an Array Object look like in Memory?



Array Object Instantiation with data

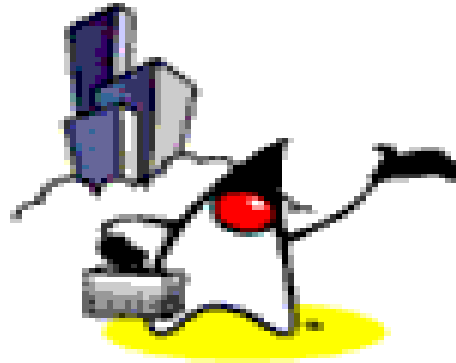
- You can also instantiate an array object by directly initializing it with data.
- For example,

```
int[] arr = {1, 2, 3, 4, 5};
```

This statement declares and instantiates an array of “int” type with five items (initialized to the values 1, 2, 3, 4, and 5).

Sample Program

```
1 // Creates an array of boolean type
2 boolean[] results = { true, false, true, false };
3 // Create an array of 4 double type initialized
4 // to the values of {100, 90, 80, 75};
5 double[] grades = {100, 90, 80, 75};
6 // Create an array of Strings
7 String[] days = { "Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"};
```



Accessing Array Element

Accessing an Array Element

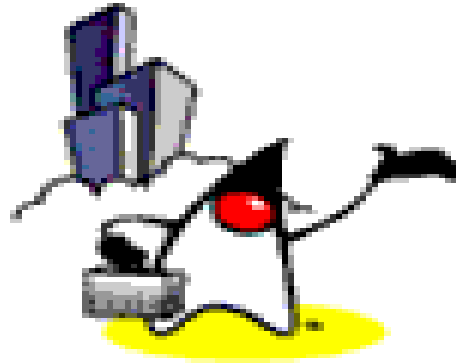
- To access an array element, you provide an **index**
- Index starts from 0

```
//assigns 10 to the third element of the array  
ages[2] = 10;
```

```
//prints the last element in the array  
System.out.println(ages[ages.length - 1]);
```

Initialization of the Array Elements

- Once an array object is created, the stored value of each member of the array will be initialized to zero for number type
 - > `int[ ] ages = new int[10];` // Each item is initialized to 0
- For reference data types such as Strings, they are initialized null
 - > `String[ ] ages = new String[10];` // Each item is initialized to null



Array Length

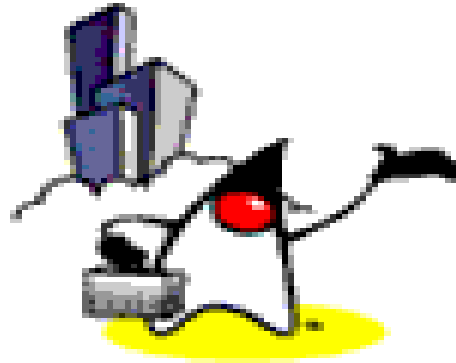
Array Length

- In order to get the number of elements in an array, you can use the length field of an array.
- The length field of an array returns the size of the array.

`arrayName.length`

Array Length

```
1  public class ArraySample {  
2      public static void main( String[] args ){  
3          int[] ages = new int[100];  
  
4          for( int i=0; i<ages.length; i++ ){  
5              System.out.print( ages[i] );  
6          }  
7      }  
8  }
```



Multi-Dimensional Array

Multidimensional Arrays

- Multidimensional arrays are implemented as arrays of arrays.
- Multidimensional arrays are declared by appending the appropriate number of bracket pairs after the array name.

Multidimensional Arrays

```
// integer array 512 x 128 elements  
int[][] twoD = new int[512][128];
```

```
// character array 8 x 16 x 24  
char[][][] threeD = new char[8][16][24];
```

```
// String array 4 rows x 2 columns  
String[][] dogs = { { "terry", "brown" },  
                    { "Kristin", "white" },  
                    { "toby", "gray"},  
                    { "fido", "black"}  
};
```

Multidimensional Arrays

- To access an element in a multidimensional array is just the same as accessing the elements in a one-dimensional array.
- For example, to access the first element in the first row of the array `dogs`, we write,

```
System.out.print( dogs[0][0] );
```

This will print the String "terry" on the screen.

Thank you!

